

Supplemental Online Materials to
Chance Constrained Programs with Gaussian Mixture Models
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1 Expectation Maximization

Suppose we have an independent and identically distributed (i.i.d.) sample from ξ , denoted as $\xi_1, \xi_2, \dots, \xi_N$. We can use the following EM algorithm (Chapter 9 of Bishop 2006) to obtain a GMM.

Expectation Maximization (EM)

Step 1. Initialize μ_j, Σ_j and $\pi_j, j = 1, \dots, K$.

Step 2. E Step. Compute

$$\gamma_{nj} = \frac{\pi_j \mathcal{N}(\xi_n | \mu_j, \Sigma_j)}{\sum_{i=1}^K \pi_i \mathcal{N}(\xi_n | \mu_i, \Sigma_i)}.$$

Step 3. M Step. Re-estimate

$$\mu_j^{\text{new}} = \frac{1}{N_j} \sum_{n=1}^N \gamma_{nj} \xi_n, \quad \Sigma_j^{\text{new}} = \frac{1}{N_j} \sum_{n=1}^N \gamma_{nj} (\xi_n - \mu_j^{\text{new}}) (\xi_n - \mu_j^{\text{new}})^\top, \quad \pi_j^{\text{new}} = \frac{N_j}{N},$$

where

$$N_j = \sum_{n=1}^N \gamma_{nj}.$$

Step 4. Check whether the convergence criterion is satisfied. If not, return to Step 2.

The convergence of the EM algorithm has been built in the literature, see, e.g., Wu (1983). In Step 4, a convergence criterion should be specified. A widely used rule in the literature is to terminate the algorithm if the difference change of the likelihood (or log-likelihood) function is smaller than a given small value (e.g., McLachlan and Peel 2000). The EM algorithm has been embedded in some softwares. For instance, in Matlab, we can directly call the EM algorithm package to conduct the estimation for the GMM and set the termination rule accordingly.

2 Gradient Formula of Henrion and Möller (2012)

Consider (5) and suppose that ξ follows $\mathcal{N}(z|\mu, \Sigma)$. Let

$$S^{(i)} = \Sigma - \frac{1}{b_i^\top \Sigma b_i} \Sigma b_i b_i^\top \Sigma, \quad w^{(i)} = \mu + \frac{x_i - b_i^\top \mu}{b_i^\top \Sigma b_i} \Sigma b_i, \quad i = 1, \dots, m. \quad (19)$$

From Lemma 3.1 of Henrion and Möller (2012), there exists a factorization

$$S^{(i)} = L^{(i)} L^{(i)\top}, \quad (20)$$

where $L^{(i)}$ is of dimension $k \times (k-1)$ and $\text{rank } L^{(i)} = k-1$.

Lemma 1 (*Theorem 3.3 of Henrion and Möller (2012)*) *Let x be such that $Bz \leq x$ is nondegenerate. Suppose that Σ is positive definite. Then*

$$\frac{\partial}{\partial x_i} \Pr \{B\xi \leq x\} = g_i(x_i) \Pr \left\{ B^{(i)} L^{(i)} \xi^{(i)} \leq x^{(i)} - B^{(i)} w^{(i)} \right\}, \quad (21)$$

where $\xi^{(i)} \sim \mathcal{N}(z|0, I_{k-1})$, $B^{(i)}$ results from B by deleting row i , $x^{(i)}$ results from x by deleting component i , $L^{(i)}$ and $w^{(i)}$ are defined in (19)-(20), and g_i is the one-dimensional Gaussian density with mean $b_i^\top \mu$ and variance $b_i^\top \Sigma b_i$. Moreover, the inequality system $B^{(i)} L^{(i)} y \leq x^{(i)} - B^{(i)} w^{(i)}$ is nondegenerate.

3 Proof of Proposition 3

Proof. By introducing an auxiliary decision vector $y = (y_1, \dots, y_K)^\top$, we can transform constraint (8) equivalently to the following set of constraints

$$\Pr_{\sim P_j} \left\{ f_0(x) + f(x)^\top \xi \leq 0 \right\} \geq y_j, \quad j = 1, \dots, K, \quad (22)$$

$$\sum_{j=1}^K \pi_j y_j \geq 1 - \alpha, \quad 0 \leq y_j \leq 1, \quad j = 1, \dots, K.$$

Since P_j is $\mathcal{N}(z|\mu_j, \Sigma_j)$, we have (22) are equivalent to (12b). This concludes the proof of the proposition. $\square$

4 Proof of Proposition 4

Proof. Because the algorithm constantly improves the lower bound by partition, the sequence $\{\underline{v}_s^i\}$ is nondecreasing. Since $\underline{v}_s^i$ is always bounded above by v_{opt} , $\underline{v}_s^i$ converge to some value v_s and $v_s \leq v_{opt}$. We next show the opposite inequality. Consider the rectangle Δ_i that generates $\underline{v}_s^i$. For simplicity we denote the lower corner of Δ_i as $\underline{y}^i$. Suppose $\{x_s^{i_h}\}_{i_h \in \mathcal{N}}$ is a subsequence of $\{x_s^i\}_{i \in \mathcal{N}}$ converging to x^* . Note that $\{\underline{y}^{i_h}\}$ is a bounded sequence. It has convergent subsequence. Thus

by passing to a subsequence if necessary, we can assume that $\{(x_s^{i_h}, \underline{y}^{i_h})\}_{i_h \in \mathcal{N}}$ converge to $(x^*, \underline{y}^*)$ for some $\underline{y}^*$. By the continuity of $h(x)$, we obtain $h(x^*) = v_s \leq v_{opt}$. We show that $(x^*, \underline{y}^*)$ is a feasible solution of Problem (12a)-(12c). First, by the partition rule in Step 2, $\underline{y}^*$ will be on the hyperplane $\sum_{j=1}^K \pi_j y_j = 1 - \alpha$.

Denote

$$F_j(x, y_j) = \Phi^{-1}(y_j) \sqrt{f(x)^\top \Sigma_j f(x)} + \left(f_0(x) + f(x)^\top \mu_j \right).$$

We have $F_j(x_s^{i_h}, \underline{y}_j^{i_h}) \leq 0$. Since $f_0(x)$, $f(x)$ and $\Phi^{-1}(y_j)$ are continuous, $F_j(x, y_j)$ is continuous. Letting $i_h \rightarrow \infty$, we have $F_j(x^*, \underline{y}_j^*) \leq 0$. This indicates $(x^*, \underline{y}^*)$ is a feasible solution of Problem (12a)-(12c). Thus $h(x^*) \geq v_{opt}$. This concludes the proof of the proposition. $\square$

5 Proof of Proposition 5

Proof. Since the region Δ in Problem (14a)-(14c) is a subset of the initial region

$$\Delta^0 := \{y \in \mathbb{R}^K \mid \underline{y}_j^0 \leq y_j \leq \bar{y}_j^0, j = 1, \dots, K\},$$

we have $\underline{y}_j \geq \underline{y}_j^0, j = 1, \dots, K$. If $\underline{y}_j^0 \geq 1/2, j = 1, \dots, K$, then $\underline{y}_j \geq 1/2, j = 1, \dots, K$. Note that $\underline{y}_j^0 = \max\{0, 1 - \alpha/\pi_j\}$. Then $\underline{y}_j^0 \geq 1/2, j = 1, \dots, K$ is equivalent to $\pi_j \geq 2\alpha, j = 1, \dots, K$. This concludes the proof. $\square$

6 Numerical Results about Comparison with SAA

Table 5: Optimal values obtained by the BB algorithm and the SAA with $\alpha = 0.05$

K	d	BB Algorithm	SAA		
		h^*	Ave	Min	Max
2	50	-4010.422	-4010.130	-4010.422	-4008.988
	80	-4975.532	N	N	N
	100	-6023.129	-6023.059	-6023.059	-6023.059
	200	-8952.419	N	N	N
3	50	-4530.938	-4530.107	-4530.938	-4526.971
	80	-5671.678	-5669.218	-5670.473	-5668.407
	100	-4786.794	N	N	N
	200	-9582.504	-8914.516	-9003.624	-8825.408
4	50	-3331.731	-3331.731	-3331.731	-3331.731
	80	-4418.467	N	N	N
	100	-6082.795	-6031.317	-6031.317	-6031.317
	200	-8110.376	-7324.050	-7324.050	-7324.050
5	50	-4527.809	-4527.361	-4527.809	-4525.771
	80	-5011.244	-5011.244	-5011.244	-5011.244
	100	-5295.994	-5290.826	-5290.826	-5290.826
	200	N	-7728.917	-7747.468	-7712.798

Table 6: Computational time of the BB algorithm and the SAA with $\alpha = 0.05$

K	d	BB Algorithm	SAA		
		Time (s)	Ave (s)	Min (s)	Max (s)
2	50	1.608	0.162	0.133	0.228
	80	3.839	38.378	10.836	66.428
	100	1.572	2564.332	905.784	3606.129
	200	16.086	3624.730	3613.804	3645.360
3	50	0.109	0.192	0.121	0.317
	80	6.120	16.971	4.537	35.537
	100	41.506	3612.926	3610.090	3617.223
	200	332.157	3619.292	3613.676	3625.294
4	50	0.068	0.143	0.111	0.237
	80	48.054	17.265	11.073	29.248
	100	824.759	3605.084	3601.586	3608.506
	200	3472.138	3616.635	3608.167	3626.797
5	50	0.065	0.183	0.114	0.354
	80	0.092	5.436	4.886	6.536
	100	403.830	805.552	66.034	3601.453
	200	3600.398	3619.643	3614.901	3622.294

Table 7: Optimal values obtained by the BB algorithm and the SAA with $\alpha = 0.1$

K	d	BB Algorithm	SAA		
		h^*	Ave	Min	Max
2	20	-1948.010	-1948.010	-1948.010	-1948.010
	30	-3021.558	-3021.558	-3021.558	-3021.558
	40	-3673.854	-3673.676	-3674.635	-3673.096
	50	-4311.580	-4316.005	-4317.489	-4314.522
3	20	-1878.896	-1878.840	-1878.896	-1878.616
	30	-2351.701	-2351.319	-2351.701	-2350.664
	40	-2869.429	-2868.893	-2869.429	-2867.026
	50	-3309.482	N	N	N
4	20	-2162.301	-2162.065	-2162.301	-2161.260
	30	-2551.568	-2551.568	-2551.568	-2551.568
	40	-2752.266	-2751.507	-2752.266	-2750.591
	50	-3576.113	-3576.113	-3576.113	-3576.113
5	20	-1724.862	-1724.541	-1724.862	-1724.041
	30	-3002.829	-3002.779	-3002.829	-3002.581
	40	-3932.573	-3932.003	-3932.573	-3931.193
	50	N	N	N	N

Table 8: Computational time of the BB algorithm and the SAA with $\alpha = 0.1$

K	d	BB Algorithm	SAA		
		Time (s)	Ave (s)	Min (s)	Max (s)
2	20	10.347	0.346	0.134	0.997
	30	27.913	0.232	0.168	0.329
	40	322.739	1.664	0.873	2.394
	50	1528.294	20.711	8.589	32.421
3	20	23.344	0.213	0.165	0.267
	30	36.402	0.195	0.154	0.232
	40	99.499	0.811	0.542	1.365
	50	3633.176	182.171	48.497	392.876
4	20	16.699	0.131	0.122	0.142
	30	32.944	0.151	0.148	0.159
	40	115.089	0.307	0.252	0.455
	50	225.416	5.640	3.204	10.717
5	20	23.220	0.170	0.115	0.232
	30	40.889	0.291	0.265	0.334
	40	120.371	1.106	0.560	1.352
	50	3690.349	375.484	32.446	780.522

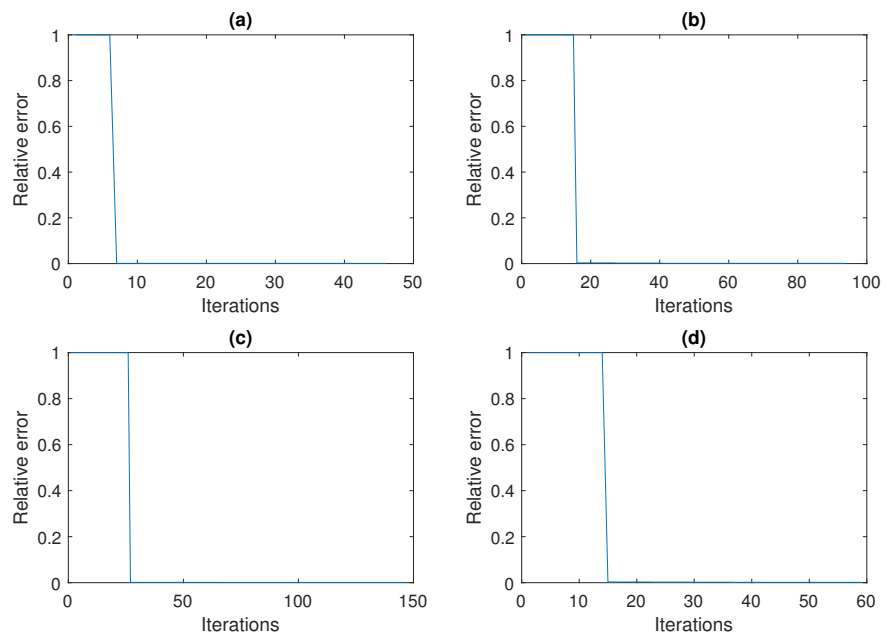


Figure 4: Performances of the BB algorithm within four replications for Problem (15a)-(15b) with $h(x) = x^\top Hx + \sum_{i=1}^d \exp(5 - x_i)$.

7 Dow Jones Credit Suisse Broad Hedge Fund Indices

Table 9: Dow Jones Credit Suisse Broad Hedge Fund indices

1	Convertible arbitrage
2	Dedicated short bias
3	Emerging markets
4	Event driven
5	Event driven distressed
6	Event driven multistrategy
7	Event driven risk arbitrage
8	Fixed income arbitrage
9	Global macro
10	Long/short equity
11	Managed futures
12	Multistrategy

Table 10: Descriptive statistics of monthly return rate of hedge fund indices

Asset	Mean(%)	Standard deviation(%)	Skewness	Kurtosis
1	0.57	1.85	-2.66	20.51
2	-0.41	4.69	0.73	4.52
3	0.62	3.87	-0.84	9.59
4	0.67	1.76	-2.07	12.67
5	0.76	1.78	-2.10	14.34
6	0.63	1.91	-1.62	9.69
7	0.48	1.14	-0.90	7.47
8	0.44	1.49	-4.80	40.29
9	0.88	2.54	0.18	7.98
10	0.74	2.64	0.02	6.94
11	0.44	3.35	0.05	2.83
12	0.63	1.43	-1.72	9.79

8 Implementation of Expectation Maximization

We use the EM algorithm to estimate the parameters of the GMM after specifying the number of components. Because too many components may lead to the badly conditioned estimated component covariances, we restrict the component number to certain range while setting the regularization value to 10^{-4} . In order to avoid inaccurate parameter estimations caused by initial values required by the EM algorithm, we design a two-stage method to fit a GMM to the underlying data as follows.

Stage 1. For each number of mixture components, fit multiple GMMs to the underlying data by repeatedly launching the EM algorithm with various starting points, and choose the GMM

with the largest loglikelihood.

Stage 2. Compare the Akaike Information Criterion (AIC) (see, e.g., Bishop 2006) among various number of mixture components, and choose the number of mixture components for which the fitted GMM is with the least AIC value.

In the first stage, we launch the EM algorithm 500 times. For each run, the EM algorithm terminates when either the maximum iteration reaches 10^4 or the loglikelihood function value meets termination tolerance 10^{-6} . We use the K-means++ algorithm for the initialization of the EM algorithm. Since K-means++ algorithm randomly selects a sample from our data as the first initial component center, the resulting starting points for the EM algorithm vary with each run. Here we only consider the number of components ranging from 1 to 10. Following the above two-stage method, we choose three components for the best fitting GMM with the least AIC statistics value as shown in Table 11.

Table 11: Comparison of AIC statistics over varying number of components

Number	1	2	3	4	5
AIC	-17567.54	-17856.76	-17919.76	-17852.36	-17780.03
Number	6	7	8	9	10
AIC	-17667.90	-17569.00	-17437.54	-17324.80	-17196.77

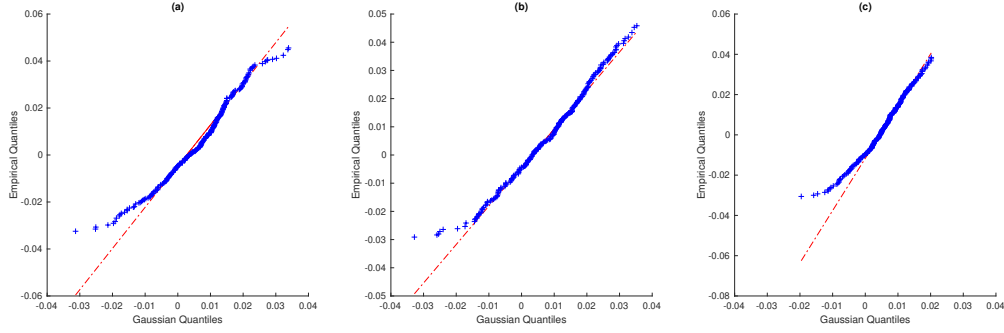


Figure 5: Quantile plots of the hedge fund return rates under Gaussian distribution: (a) Convertible arbitrage; (b) Event driven distressed; (c) Fixed income arbitrage

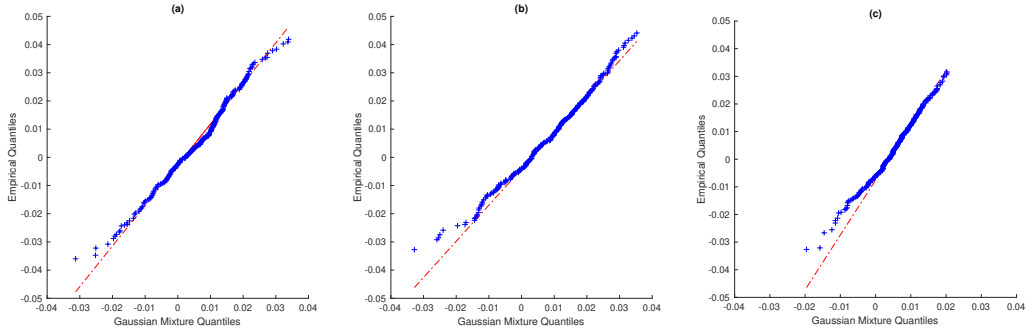


Figure 6: Quantile plots of the hedge fund return rates under Gaussian mixture model: (a) Convertible arbitrage; (b) Event driven distressed; (c) Fixed income arbitrage

9 Out-of-Sample Performances

We further compare the out-of-sample performances of the optimal portfolios generated from the GMM and the Gaussian distribution fitted by the same empirical return rate data. We divide the whole data set into the in-sample and out-of-sample parts. More specifically, the initial in-sample data set consists of the monthly returns of the 12 indices from April 1994 to December 2011 with the remaining part being the initial out-of-sample data. The in-sample data are used to estimate the parameters of the GMM and Gaussian distribution with the EM algorithm. The out-of-sample data are used to compare the performances of different portfolios constructed by solving the Problem (17a)-(17c) with the fitted GMM and the Gaussian distribution. More specifically, we conduct the out-of-sample analysis of the constructed portfolios using a back testing strategy by calculating performance criteria of different portfolios with the monthly returns in out-of-sample period. We gradually increase the sample size of the in-sample data set with the most recent three monthly

return rate data and repeatedly solve Problem (17a)-(17c) every three months to generate the monthly portfolio rebalance strategies based on the newly fitted GMM and Gaussian distribution.

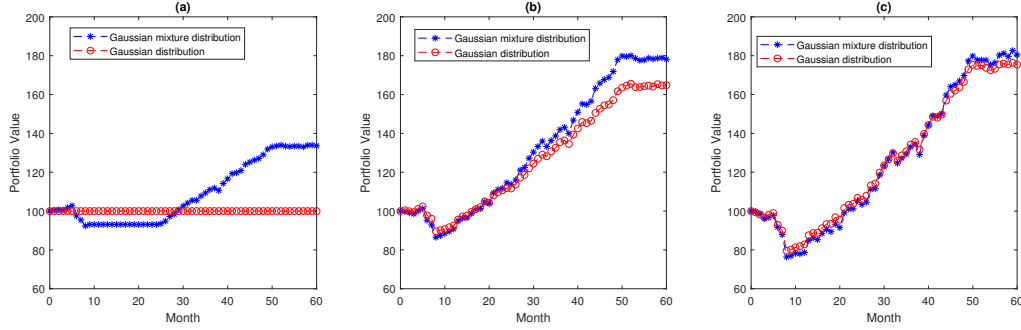


Figure 7: Evolution of portfolio values during out-of-sample period for $\alpha = 0.05$ under different settings: GMM versus Gaussian distribution when (a) $v = 0.01$, (b) $v = 0.015$ and (c) $v = 0.02$.

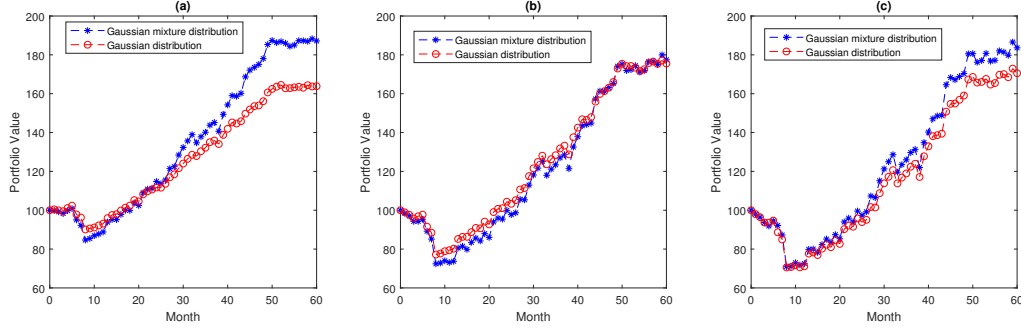


Figure 8: Evolution of portfolio values during out-of-sample period for $\alpha = 0.1$ under different settings: GMM versus Gaussian distribution when (a) $v = 0.01$, (b) $v = 0.015$ and (c) $v = 0.02$.

Figures 7 and 8 illustrate the evolution of the values of different portfolios constructed by the GMM and Gaussian distribution under different combinations of v and α using the monthly rebalancing strategy. As shown in the figures, given the same initial portfolio value 100, the optimal portfolios derived from the GMM can often achieve better portfolio values than using a Gaussian distribution in the long run. The GMM based model can performs well in the case of $v = 0.01$, which may suggest that it is more suitable for extreme cases than the Gaussian model. Note that when $v = 0.01$ and $\alpha = 0.05$, there are no feasible portfolios under the Gaussian distribution. However, from the figures we find that the volatility of the optimal portfolios derived from the GMM is larger than that from the Gaussian distribution. Tables 12 and 13 show respectively the out-of-sample statistical results of the optimal portfolios generated by the GMM and Gaussian distribution, including the final portfolio value ($Finval$), average return rate (R_{Ave}), maximum return rate (R_{Max}), minimum return rate (R_{Min}), standard deviation (Std), and the ratio of average return rate over standard deviation (R_{Ave}/Std) of monthly return rate. In accordance with the

evolution of the portfolio values in Figures 7 and 8, Tables 12 and 13 suggest that the portfolios derived from the GMM often show a better performance than that from the Gaussian distribution in terms of the final portfolio values and the average return rate under various combinations of v and α . However, as reflected in Figures 7 and 8, the standard deviations in Tables 12 and 13 show that the volatility of portfolios generated by the GMM is larger than that by the Gaussian distribution under all combinations of v and α . Thus, in terms of the ratio of average return rate over standard deviation, GMM does not enjoy a superiority over the Gaussian distribution.

Table 12: Statistical results of portfolio values and return rates for the Gaussian distribution during the out-of-sample period

α	v	Gaussian distribution					
		Finval	$R_{Ave}(\%)$	$R_{Min}(\%)$	$R_{Max}(\%)$	$Std(\%)$	$R_{Ave}/Std(\%)$
0.05	0.01	100.000	N	N	N	N	N
	0.015	164.823	0.851	-6.768	3.676	1.734	0.491
	0.02	175.422	0.981	-11.505	6.256	2.812	0.349
0.1	0.01	163.873	0.840	-6.377	3.486	1.666	0.505
	0.015	175.531	0.990	-12.697	6.839	3.092	0.320
	0.02	170.550	0.981	-16.924	9.216	4.174	0.235

Table 13: Statistical results of portfolio values and return rates for the GMM during the out-of-sample period

α	v	Gaussian mixture model					
		Finval	$R_{Ave}(\%)$	$R_{Min}(\%)$	$R_{Max}(\%)$	$Std(\%)$	$R_{Ave}/Std(\%)$
0.05	0.01	133.622	0.657	-5.096	3.261	1.490	0.441
	0.015	178.169	0.991	-6.849	5.059	2.179	0.455
	0.02	180.437	1.049	-13.147	8.377	3.485	0.301
0.1	0.01	187.178	1.083	-8.206	6.389	2.583	0.419
	0.015	177.446	1.044	-14.963	9.261	4.086	0.255
	0.02	183.633	1.131	-18.668	10.600	4.740	0.239

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